

Intro to Analysis

MATH 147

Qinghao Hu

September 26, 2026

Contents

1	Course Information	3
1.1	Logistics	3
1.1.1	Contact	3
1.1.2	Course Structure	3
2	The Real Number System	4
2.1	Numbers and Logic	4
2.1.1	What Is Calculus?	4
2.1.2	From $\mathbb{N}$ to $\mathbb{Q}$	4
2.1.3	Statements and Quantifiers	5
2.2	Induction and Well-Ordering	7
2.2.1	Mathematical Induction	7
2.2.2	The Well-Ordering Principle	9
2.3	$\mathbb{R}$ as an Ordered Field	12
2.3.1	The Field Axioms	12
2.3.2	The Order Axioms	14
2.3.3	Absolute Value and the Triangle Inequality	16
2.4	Suprema and Infima	19
2.4.1	The ϵ -Characterization of the Supremum	20
2.5	Completeness and Its Consequences	22
2.5.1	The Archimedean Property	23
2.5.2	The Existence of $\sqrt{2}$	26
2.5.3	Density of $\mathbb{Q}$ and of the Irrationals	29
2.5.4	Countable and Uncountable Sets	32
3	Sequences and Limits	33
3.1	Convergent Sequences	33
3.1.1	The Limit of a Sequence	33
3.1.2	Uniqueness of the Limit	34
3.1.3	Divergence and Boundedness	35
3.1.4	Divergence to $\pm\infty$	37

1 Course Information

1.1 Logistics

1.1.1 Contact

Email: mmolino@uwaterloo.ca

Website: www.math.uwaterloo.ca/~mmolino

Office hours: Monday, 3:30–4:30 p.m., and Friday, 10:00–11:00 a.m., in MC 4006.

1.1.2 Course Structure

Lectures: Section 002 meets on Monday, Wednesday, and Friday from 12:30–1:20 p.m. in MC 4021.

2 The Real Number System

2.1 Numbers and Logic

2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

2.1.2 From $\mathbb{N}$ to $\mathbb{Q}$

The number systems discussed in this lecture include

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

$$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\},$$

$$\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1.1 (Irrationality of $\sqrt{2}$)

The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.

In symbols

$$\neg(\exists m \in \mathbb{Z})(\exists n \in \mathbb{N}) \left[\sqrt{2} = \frac{m}{n} \right]$$

Idea

Write $\sqrt{2} = m/n$ in lowest terms. Then $m^2 = 2n^2$ forces m even, and feeding that back forces n even — so 2 divides both, contradicting lowest terms.

Proof

Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 2n^2.$$

Thus m^2 is even. An odd integer has an odd square, so m must be even. Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned} 2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2. \end{aligned}$$

Therefore n^2 is even, and the same parity argument shows that n is even.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

Assuming lowest terms turns the equation into a parity contradiction.

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.1.2 (Decimal representation)

Every nonnegative real number has a decimal representation of the form

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Negative real numbers are the negatives of such expansions. Some numbers have two decimal representations, for example $1.999 \dots = 2.000 \dots$.

Example 2.1.3 (Recurring decimal)

Is the statement $1.999 \dots = 2$ correct?

Yes. Let $x = 1.999 \dots$. Then

$$\begin{aligned} x &= 1.999 \dots, \\ 10x &= 19.999 \dots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

2.1.3 Statements and Quantifiers

Definition 2.1.4 (Statement)

A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 2.1.5

The statement

$$(\forall x \in \mathbb{R}) [x^2 - x \geq 0]$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 2.1.6

The statement

$$(\exists x \in \mathbb{R}) [x^2 - x \geq 0]$$

is true; for instance, $x = 2$ satisfies the inequality.

Notation

Each quantifier is written in parentheses, and the statement it governs in square brackets, read left to right:

$$(\forall x \in \mathbb{R})(\exists y \in \mathbb{R}) [x^3 - y^3 = 1]$$

reads “for every $x \in \mathbb{R}$ there exists $y \in \mathbb{R}$ such that $x^3 - y^3 = 1$.” Negation moves inward and swaps each quantifier:

$$\neg(\forall x)(\exists y) [P(x, y)] \iff (\exists x)(\forall y) [\neg P(x, y)].$$

From here on, every theorem-like statement ends with its form *In symbols*: quantifiers and connectives only, using notation defined before that statement.

Proposition 2.1.7

For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that

$$x^3 - y^3 = 1.$$

In symbols

$$(\forall x \in \mathbb{R})(\exists y \in \mathbb{R}) [x^3 - y^3 = 1]$$

Proof

Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Here the choice of y is allowed to depend on x : for each x we produce a (possibly different) y that makes the equation hold. This is exactly what the quantifier order $(\forall x)(\exists y)$ permits.

Proposition 2.1.8

The statement

$$(\exists y \in \mathbb{R})(\forall x \in \mathbb{R}) [x^3 - y^3 = 1]$$

is false.

In symbols

$$\neg(\exists y \in \mathbb{R})(\forall x \in \mathbb{R}) [x^3 - y^3 = 1]$$

Its negation is

$$(\forall y \in \mathbb{R})(\exists x \in \mathbb{R}) [x^3 - y^3 \neq 1].$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark

In this second statement, the quantifier order is reversed: $(\exists y)(\forall x)$ demands a *single* y that works simultaneously for every $x \in \mathbb{R}$. This is a much stronger requirement than $(\forall x)(\exists y)$, which only asks that a (possibly different) y exist for each individual x . The first proposition shows such y 's exist, but they genuinely depend on x (namely $y = \sqrt[3]{x^3 - 1}$), so no single choice of y can work for all x at once. This is why swapping the order of $\forall$ and $\exists$ can change a true statement into a false one, and it is a caution to always read quantified statements in the order they are written.

2.2 Induction and Well-Ordering**2.2.1 Mathematical Induction****Definition 2.2.1** (Principle of Mathematical Induction)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true;

then

$$(\forall n \in \mathbb{N}) [P(n)]$$

is true.

In symbols

$$\left(P(1) \wedge (\forall k \in \mathbb{N}) [P(k) \Rightarrow P(k + 1)] \right) \implies (\forall n \in \mathbb{N}) [P(n)]$$

Here is an example:

Proposition 2.2.2 (Sum of the First n Natural Numbers)

For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

In symbols

$$(\forall n \in \mathbb{N}) \left[\sum_{k=1}^n k = \frac{n(n+1)}{2} \right]$$

Proof

Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1+1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, $P(k+1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 2.2.3

Let $S \subseteq \mathbb{N}$. If

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k+1 \in S$;

then $S = \mathbb{N}$.

In symbols

$$(\forall S \subseteq \mathbb{N}) \left[(1 \in S \wedge (\forall k \in S) [k+1 \in S]) \Rightarrow S = \mathbb{N} \right]$$

Proof

Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k+1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

2.2.2 The Well-Ordering Principle**Theorem 2.2.4** (Well-Ordering Principle)

Every nonempty subset S of $\mathbb{N}$ has a smallest element.

In symbols

$$(\forall S \subseteq \mathbb{N}) \left[S \neq \emptyset \Rightarrow (\exists m \in S)(\forall s \in S) [m \leq s] \right]$$

Idea

If S had no smallest element, induction would show that S misses all of $1, 2, \dots, n$ for every n — so $S = \emptyset$.

Proof

Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

That is, T consists of exactly those $n \in \mathbb{N}$ for which none of $1, 2, \dots, n$ lies in S — informally, $n \in T$ means S has "not yet been seen" by the time we reach n . If we can show $T = \mathbb{N}$, then in particular no natural number lies in S (since every $n \in \mathbb{N}$ would satisfy $n \in T$, forcing $n \notin S$), giving $S = \emptyset$ and the desired contradiction. We show $T = \mathbb{N}$ by induction on n .

Base case. Since S has no smallest element and 1 is the smallest element of $\mathbb{N}$, we must have $1 \notin S$ (otherwise 1 would be the smallest element of S). Hence $\{1\} \cap S = \emptyset$, so $1 \in T$.

Inductive step. Suppose $k \in T$, i.e., $\{1, 2, \dots, k\} \cap S = \emptyset$. We show $k+1 \in T$. Suppose, for contradiction, that $k+1 \notin T$; then $\{1, 2, \dots, k+1\} \cap S \neq \emptyset$. Since $\{1, \dots, k\} \cap S = \emptyset$ by the inductive hypothesis, the only way for $\{1, \dots, k+1\} \cap S$ to be nonempty is if $k+1 \in S$. But then $k+1 \in S$ while none of $1, 2, \dots, k$ lie in S , so $k+1$ would be the smallest element of S — contradicting our assumption that S has no smallest element. Hence $k+1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so $S = \emptyset$, contradicting the assumption that S is nonempty.

Therefore, S has a smallest element. ■

Proposition 2.2.5

Using the well-ordering principle, prove that for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

In symbols

$$(\forall n \in \mathbb{N}) \left[\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \right]$$

Idea

Minimal counterexample: let k be the first n where the formula fails. The formula still holds at $k-1$, and adding k^2 carries it to k .

Proof

Let $P(n)$ denote the statement

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Suppose, for contradiction, that $P(n)$ fails for some n ; that is,

$$\neg(\forall n \in \mathbb{N}) [P(n)] \quad \text{equivalently} \quad (\exists n \in \mathbb{N}) [\neg P(n)].$$

Then the set

$$S := \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By the well-ordering principle, S has a smallest element k . Since $P(1)$ holds, $1 \notin S$, so $k > 1$ and hence $k-1 \in \mathbb{N}$.

By minimality of k , we have $k-1 \notin S$, so $P(k-1)$ is true:

$$1^2 + 2^2 + \cdots + (k-1)^2 = \frac{(k-1)k(2(k-1)+1)}{6} = \frac{(k-1)k(2k-1)}{6}.$$

Adding k^2 to both sides gives

$$\begin{aligned} 1^2 + \cdots + (k-1)^2 + k^2 &= \frac{(k-1)k(2k-1)}{6} + k^2 \\ &= \frac{(k-1)k(2k-1) + 6k^2}{6} \\ &= \frac{k((k-1)(2k-1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k+1)(2k+1)}{6}, \end{aligned}$$

so $P(k)$ is in fact true. This contradicts $k \in S$. Therefore $S = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$. ■

Remark (Why min, not sup)

It is worth pausing on why we take the *minimum* of S here rather than some supremum-type argument. The well-ordering principle applies specifically to nonempty subsets of $\mathbb{N}$ (not to general subsets of $\mathbb{R}$), and it guarantees the existence of a smallest element that actually *belongs to* S — this is what allows us to write $k = \min S \in S$ and conclude $\neg P(k)$. This is a stronger statement than merely having an infimum: for a general subset of $\mathbb{R}$, an infimum need not belong to the set, so we could not conclude $\neg P(\inf S)$. Here, because $S \subseteq \mathbb{N}$ and S is nonempty, well-ordering hands us an actual least element k with $\neg P(k)$ true, which is exactly what is needed to test P at k and at $k-1$.

It is also worth noting that induction and the well-ordering principle are logically equivalent ways of expressing the same underlying fact about $\mathbb{N}$: the proof of the well-ordering principle above used induction, and here we use well-ordering to reprove a statement ($P(n)$ for all n) that could equally well have been proved by direct induction on n . The two techniques are simply different packagings of the same idea.

In practice

- **Induction.** To prove $P(n)$ for every $n \in \mathbb{N}$: verify the *base case* $P(1)$ directly; then fix an arbitrary $k \in \mathbb{N}$, assume $P(k)$ (the *inductive hypothesis*), and deduce $P(k + 1)$. No individual case needs to be checked beyond these two steps.
- **Minimal counterexample.** Suppose $P(n)$ fails somewhere. The counterexamples $S = \{n \in \mathbb{N} : \neg P(n)\}$ form a nonempty subset of $\mathbb{N}$, so they have a least element k . Because k is the *first* failure, $P(k - 1)$ still holds; show that $P(k - 1)$ forces $P(k)$:

$$S \neq \emptyset \Rightarrow k = \min S \Rightarrow P(k - 1) \Rightarrow P(k) \Rightarrow k \notin S, \text{ contradiction.}$$

2.3 $\mathbb{R}$ as an Ordered Field

The real numbers $\mathbb{R}$ form a *complete ordered field*. We build up this description in three stages: first the field (algebraic) axioms, which govern $+$ and $\cdot$; then the order axioms, which let us compare elements with $<$; and finally the completeness axiom, which is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is the source of all limiting behaviour in analysis. Each stage adds structure on top of the previous one, and each is needed for a different purpose: the field axioms let us compute, the order axioms let us compare, and completeness lets us take limits.

2.3.1 The Field Axioms

Definition 2.3.1 (Field)

A **field** is a set F equipped with two binary operations

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F,$$

satisfying the following axioms for all $a, b, c \in F$.

F1. Associativity of addition.

$$a + (b + c) = (a + b) + c.$$

F2. Commutativity of addition.

$$a + b = b + a.$$

F3. Additive identity. There exists an element $0 \in F$ such that

$$a + 0 = a.$$

F4. Additive inverses. For every $a \in F$, there exists an element $-a \in F$ such that

$$a + (-a) = 0.$$

F5. Associativity of multiplication.

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

F6. Commutativity of multiplication.

$$a \cdot b = b \cdot a.$$

F7. Multiplicative identity. There exists an element $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

F8. Multiplicative inverses. For every $a \in F$ with $a \neq 0$, there exists an element $a^{-1} \in F$ such that

$$a \cdot a^{-1} = 1.$$

F9. Distributive law.

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

The rational numbers $\mathbb{Q}$ also satisfy F1–F9, so being a field does not distinguish $\mathbb{R}$ from $\mathbb{Q}$: the field axioms capture the algebra of $+$ and $\cdot$, but nothing about size, order, or completeness.

Once the axioms are fixed, even the most familiar facts — uniqueness of the identities, $a \cdot 0 = 0$ — become theorems.

Theorem 2.3.2 a) If $z, a \in \mathbb{R}$ with $z + a = a$, then $z = 0$.

b) If $u \in \mathbb{R}$ and $b \neq 0$ with $u \cdot b = b$, then $u = 1$.

c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

In symbols

a) $(\forall z, a \in \mathbb{R}) [z + a = a \Rightarrow z = 0]$

b) $(\forall u, b \in \mathbb{R}) [(b \neq 0 \wedge u \cdot b = b) \Rightarrow u = 1]$

c) $(\forall a \in \mathbb{R}) [a \cdot 0 = 0]$

Proof

a)

$$z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0.$$

b) Multiplying $u \cdot b = b$ by b^{-1} ,

$$u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1.$$

c)

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides, and using associativity, gives $0 = a \cdot 0$. ■

2.3.2 The Order Axioms

Definition 2.3.3 (The Order Properties of $\mathbb{R}$)

There is a nonempty subset P of $\mathbb{R}$, called the set of **positive real numbers**, that satisfies the following properties.

O1. Trichotomy law. For every $a \in \mathbb{R}$, exactly one of the following holds:

$$a = 0, \quad a \in P, \quad -a \in P.$$

O2. Closure under addition. If $a, b \in P$, then $a + b \in P$.

O3. Closure under multiplication. If $a, b \in P$, then $a \cdot b \in P$.

Definition 2.3.4 (Order relation)

Let $a, b \in \mathbb{R}$.

(i) If $a - b \in P$, then we write $a > b$ or $b < a$.

(ii) If $a - b \in P \cup \{0\}$, then we write $a \geq b$ or $b \leq a$.

The positive rationals $P \cap \mathbb{Q}$ also satisfy O1–O3, so an ordered field still need not be $\mathbb{R}$. The missing ingredient is completeness, which $\mathbb{Q}$ lacks: the set

$$\{q \in \mathbb{Q} : q^2 < 2\}$$

is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$. That is why completeness is a separate axiom.

Definition 2.3.5 (Completeness Axiom)

If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

In symbols

$$\begin{aligned} & (\forall A \subseteq \mathbb{R}) \left[(A \neq \emptyset \wedge (\exists u \in \mathbb{R})(\forall a \in A) [a \leq u]) \right. \\ & \quad \Rightarrow (\exists u_0 \in \mathbb{R}) [(\forall a \in A) [a \leq u_0] \\ & \quad \quad \left. \wedge (\forall u \in \mathbb{R})(\forall a \in A) [a \leq u] \Rightarrow u_0 \leq u] \right] \end{aligned}$$

This is only a preview: upper bounds, suprema and the consequences of this axiom are developed in §5.

Example 2.3.6

Prove that if $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof

Since $a \neq 0$, the trichotomy law gives two cases.

Case 1: $a \in P$. By closure under multiplication,

$$a^2 = a \cdot a \in P, \quad \text{so } a^2 > 0.$$

Case 2: $a \notin P$. Since $a \neq 0$, trichotomy rules out $a = 0$, so it must be that $-a \in P$. By closure under multiplication, $(-a)(-a) \in P$. It remains to show that $a^2 = (-a)(-a)$, which we do by starting from $a + (-a) = 0$ and multiplying both sides by $(-a)$:

$$\begin{aligned} a + (-a) = 0 & \implies (-a)(a + (-a)) = (-a) \cdot 0 = 0 \\ & \implies (-a)(a) + (-a)(-a) = 0 \\ & \implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

Hence $a^2 = (-a)(-a) \in P$, so $a^2 > 0$. Since the two cases $a \in P$ and $-a \in P$

are exhaustive (trichotomy, together with $a \neq 0$), we conclude $a^2 > 0$ in either case. ■

2.3.3 Absolute Value and the Triangle Inequality

Definition 2.3.7

Given $a \in \mathbb{R}$, we define the **absolute value** $|a|$ by

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Geometrically, $|a|$ is the distance from a to 0, and $|a - b|$ is the distance between a and b ; the results below are statements about distances.

Theorem 2.3.8 (a) $|ab| = |a| |b|$ for all $a, b \in \mathbb{R}$.

(b) $|a|^2 = a^2$ for all $a \in \mathbb{R}$.

(c) For $a \in \mathbb{R}$ and $c > 0$,

$$|a| \leq c \iff -c \leq a \leq c.$$

(d) For every $a \in \mathbb{R}$,

$$-|a| \leq a \leq |a|.$$

In symbols

$$(a) \quad (\forall a, b \in \mathbb{R}) \left[|ab| = |a| |b| \right]$$

$$(b) \quad (\forall a \in \mathbb{R}) \left[|a|^2 = a^2 \right]$$

$$(c) \quad (\forall a \in \mathbb{R}) (\forall c > 0) \left[|a| \leq c \iff -c \leq a \leq c \right]$$

$$(d) \quad (\forall a \in \mathbb{R}) \left[-|a| \leq a \leq |a| \right]$$

Idea

Every part is the same move: split on the sign of a (and of b), and unwind the definition of $|\cdot|$ in each case.

Proof

The definition immediately gives $|-a| = |a|$: if $a \geq 0$, then $-a \leq 0$, and if $a < 0$, then $-a > 0$. It also gives

$$-|a| \leq a \leq |a|,$$

proving (d). In the two cases $a \geq 0$ and $a < 0$, respectively, $|a|^2 = a^2$ is immediate; this proves (b).

For (a), there are four sign combinations. If $a, b \geq 0$, then $ab \geq 0$, so $|ab| = ab = |a| |b|$. If $a \geq 0 > b$, then $ab \leq 0$, so $|ab| = -ab = a(-b) = |a| |b|$. The case $b \geq 0 > a$ is the same with the roles reversed. Finally, if $a, b < 0$, then $ab > 0$, and

$$|ab| = ab = (-a)(-b) = |a| |b|.$$

Finally let $c > 0$. Suppose first that $|a| \leq c$. If $a \geq 0$, then $|a| = a$, so $a \leq c$; also $-c < 0 \leq a$. If $a < 0$, then $|a| = -a$, so $-a \leq c$, equivalently $-c \leq a$; also $a < 0 < c$. Hence $-c \leq a \leq c$ in either case.

Conversely, suppose $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$. If $a < 0$, multiplying $-c \leq a$ by -1 reverses the inequality and gives $-a \leq c$, hence $|a| \leq c$. This proves (c). ■

Proposition 2.3.9 (Triangle Inequality)

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

In symbols

$$(\forall a, b \in \mathbb{R}) [|a + b| \leq |a| + |b|]$$

Idea

Add the two-sided bounds from part (d) for a and for b ; this gives $-c \leq a+b \leq c$ with $c = |a| + |b|$, and part (c) turns that back into $|a + b| \leq c$.

Proof

Let $a, b \in \mathbb{R}$. By part (d) of the preceding theorem,

$$\begin{aligned} -|a| &\leq a \leq |a|, \\ -|b| &\leq b \leq |b|. \end{aligned}$$

Adding these two chains of inequalities termwise gives

$$\begin{aligned} -|a| - |b| &\leq a + b \leq |a| + |b| \\ \iff -(|a| + |b|) &\leq a + b \leq |a| + |b|. \end{aligned}$$

If $|a| + |b| = 0$, then the displayed bounds force $a + b = 0$, so the claim is immediate. Otherwise $|a| + |b| > 0$, and applying part (c) of the preceding theorem with $x = a + b$ and $c = |a| + |b|$ converts this two-sided bound back

into the single inequality

$$|a + b| \leq |a| + |b|.$$

■

In terms of distance: going from 0 to $a + b$ directly is never longer than going via a .

Corollary 2.3.10 (i) $||a| - |b|| \leq |a - b|$.

(ii) $|a - b| \leq |a| + |b|$

In symbols

$$(i) \quad (\forall a, b \in \mathbb{R}) \left[||a| - |b|| \leq |a - b| \right]$$

$$(ii) \quad (\forall a, b \in \mathbb{R}) \left[|a - b| \leq |a| + |b| \right]$$

Proof of (1)

Let $a, b \in \mathbb{R}$. The idea is to write a as $(a - b) + b$, so that the triangle inequality just proved applies directly to this sum and produces a bound involving $|a - b|$:

$$\begin{aligned} a = (a - b) + b &\implies |a| = |(a - b) + b| \leq |a - b| + |b| \\ &\implies |a| - |b| \leq |a - b|. \end{aligned}$$

Symmetrically, writing b as $(b - a) + a$ and applying the triangle inequality the same way gives

$$\begin{aligned} b = (b - a) + a &\implies |b| = |(b - a) + a| \leq |b - a| + |a| \\ &\implies |b| - |a| \leq |b - a| = |-(a - b)| = |a - b| \\ &\implies |a| - |b| \geq -|a - b|. \end{aligned}$$

Combining the two bounds obtained from these two decompositions,

$$-|a - b| \leq |a| - |b| \leq |a - b|.$$

By part (c) of the absolute value theorem, applied with $x = |a| - |b|$ and $c = |a - b|$, this two-sided bound is equivalent to

$$||a| - |b|| \leq |a - b|.$$

■

Proof of (2)

Let $a, b \in \mathbb{R}$. Writing $a - b$ as $a + (-b)$ reduces this to a direct application of

the triangle inequality, using $|-b| = |b|$:

$$|a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

■

In practice

- **Algebra is F1–F9.** Every “obvious” manipulation is an axiom in disguise: moving a term across “=” uses additive inverses and associativity, cancelling a nonzero factor uses multiplicative inverses, expanding a product uses the distributive law.
- **Case-split with trichotomy.** Any $a \in \mathbb{R}$ falls in exactly one of $a = 0$, $a \in P$, $-a \in P$; handle each case, and use O2, O3 to keep positivity through the algebra (as in the proof that $a \neq 0 \Rightarrow a^2 > 0$).
- **To show $a > b$, show $a - b \in P$.** Every order manipulation (adding to both sides, multiplying by a positive number) traces back to this definition and O1–O3.
- **Unpack absolute values.** $|x| \leq c \iff -c \leq x \leq c$ turns one absolute-value inequality into two ordinary ones, and back. The triangle inequality is the first example; it will be used constantly.
- **Add and subtract.** To bound $|a|$ in terms of $|a - b|$, write $a = (a - b) + b$ and apply the triangle inequality.

2.4 Suprema and Infima

Bounds make sense in any ordered set. The special feature of $\mathbb{R}$ is that every nonempty set bounded above has a supremum in $\mathbb{R}$; this need not hold in $\mathbb{Q}$.

Definition 2.4.1 (Bounded Sets)

Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be

- (i) bounded above if there exists $u \in \mathbb{R}$ such that $x \leq u$ for all $x \in S$.
- (ii) bounded below if there exists $w \in \mathbb{R}$ such that $w \leq x$ for all $x \in S$.
- (iii) bounded if it is both bounded above and below.
- (iv) unbounded if it is not bounded.

Definition 2.4.2 (Supremum)

If there exists an upper bound u_0 of S such that whenever u is an upper bound for S we have $u_0 \leq u$, then u_0 is called the least upper bound, or the

supremum, of S .

$$u_0 := \sup S.$$

Definition 2.4.3 (Infimum)

If there exists a lower bound w_0 of S such that whenever w is a lower bound for S we have $w_0 \geq w$, then w_0 is called the greatest lower bound, or the infimum, of S .

$$w_0 := \inf S.$$

A set has at most one supremum. If u_1 and u_2 are both suprema of S , then u_1 is least and u_2 is an upper bound, so $u_1 \leq u_2$; swapping roles gives $u_2 \leq u_1$. Hence $u_1 = u_2$, and the same argument with inequalities reversed shows the infimum is unique.

2.4.1 The ϵ -Characterization of the Supremum

When does an upper bound u fail to be the *least* one? Exactly when some smaller number is still an upper bound — that is, when there is a gap between S and u . So u is the supremum precisely when there is no gap: however small a margin $\epsilon > 0$ we allow, S still has a point within ϵ of u .

Lemma 2.4.4

An upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S) [u - \epsilon < x_\epsilon].$$

In symbols

$$\begin{aligned} & (\forall S \subseteq \mathbb{R})(\forall u \in \mathbb{R}) \left[(S \neq \emptyset \wedge (\forall x \in S) [x \leq u]) \right. \\ & \quad \left. \Rightarrow (u = \sup S \iff (\forall \epsilon > 0)(\exists x_\epsilon \in S) [u - \epsilon < x_\epsilon]) \right] \end{aligned}$$

Proof

($\Rightarrow$) Suppose $u = \sup S$. Let $\epsilon > 0$. Since

$$u - \epsilon < u,$$

the number $u - \epsilon$ cannot be an upper bound: otherwise it would be an upper bound smaller than u , contradicting that u is the *least* upper bound. Since $u - \epsilon$ is not an upper bound, by the definition of upper bound there is some $x_\epsilon \in S$

such that

$$u - \epsilon < x_\epsilon \leq u.$$

($\Leftarrow$) Conversely, suppose u is an upper bound and that for every $\epsilon > 0$ there exists $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$. If u were not the least upper bound, there would be another upper bound $w < u$. Choose

$$\epsilon = u - w > 0.$$

The assumed condition then yields some $x_\epsilon \in S$ with

$$w = u - \epsilon < x_\epsilon,$$

contradicting that w is an upper bound (since $x_\epsilon \in S$ exceeds w). Hence no such w exists, and $u = \sup S$. ■

Example 2.4.5

Consider the set $S = \{x \in \mathbb{R} : 0 \leq x < 1\}$. Then $\sup S = 1$ and $\inf S = 0$.

Proof that $\sup S = 1$

Every $x \in S$ satisfies $x < 1$, so 1 is an upper bound. Let $u = \sup S$, which exists by completeness since S is nonempty and bounded above. Because 1 is an upper bound, $u \leq 1$. Suppose $u \neq 1$, so $u < 1$. Choose

$$\epsilon = \frac{1 - u}{2} > 0.$$

Then

$$u < u + \epsilon = \frac{u + 1}{2} < 1.$$

Also $u \geq 0$ since $0 \in S$, so $u + \epsilon \geq 0$. Thus $u + \epsilon \in S$, contradicting that u is an upper bound for S . Therefore $u = 1$, i.e. $\sup S = 1$. ■

Proof that $\inf S = 0$

Since $x \geq 0$ for all $x \in S$, 0 is a lower bound. Let $w = \inf S$. Because 0 is a lower bound, $w \geq 0$. Suppose $w \neq 0$, so $w > 0$. But $0 \in S$, and a lower bound w must satisfy $w \leq 0$, a contradiction. Therefore $w = 0$, i.e. $\inf S = 0$. ■

Remark

In this example, $S = [0, 1)$ has $\sup S = 1$ but no maximum, because $1 \notin S$. Its infimum 0 does belong to S , so 0 is also its minimum. A supremum is the *least*

upper bound, whether or not it belongs to the set; it is a maximum only when it does belong to the set. Similarly, “bounded above,” “bounded below,” “bounded,” and “unbounded” are properties of the *set* S , whereas $\sup S$ and $\inf S$ (when they exist) are specific *real numbers* associated to S .

In practice

- **Verifying** $u = \sup S$. Check two things: (i) u is an upper bound for S ; (ii) no smaller number is. Condition (ii) quantifies over all numbers below u and is awkward directly; use the ϵ -form instead: for each $\epsilon > 0$, exhibit $x \in S$ with $u - \epsilon < x \leq u$.
- **Infimum**. The same recipe with inequalities reversed: a lower bound w , and for each $\epsilon > 0$ some $x \in S$ with $w \leq x < w + \epsilon$.

2.5 Completeness and Its Consequences

Axiom (Completeness Property)

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

In symbols

$$(\forall S \subseteq \mathbb{R}) \left[(S \neq \emptyset \wedge (\exists u \in \mathbb{R})(\forall x \in S) [x \leq u]) \right. \\ \left. \Rightarrow (\exists u_0 \in \mathbb{R}) [u_0 = \sup S] \right]$$

Corollary 2.5.1

Every nonempty set of real numbers that has a lower bound also has an infimum in $\mathbb{R}$.

In symbols

$$(\forall S \subseteq \mathbb{R}) \left[(S \neq \emptyset \wedge (\exists w \in \mathbb{R})(\forall x \in S) [w \leq x]) \right. \\ \left. \Rightarrow (\exists w_0 \in \mathbb{R}) [w_0 = \inf S] \right]$$

Proof

Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , that is, $w \leq s$ for all $s \in S$. As a consequence, $-w \geq -s$ for all $s \in S$.

Define $S' = \{-s : s \in S\}$. Then S' is nonempty, and

$$(\forall s \in S) [-w \geq -s] \implies -w \text{ is an upper bound for } S'.$$

By the Completeness Axiom, S' has a supremum. Say

$$u := \sup S' \in \mathbb{R}.$$

We claim that $-u = \inf S$.

- **Lower bound:** Since $u = \sup S' \geq -s$ for all $s \in S$, we get $-u \leq s$ for all $s \in S$. So $-u$ is a lower bound for S .
- **Greatest lower bound:** Let w be any lower bound for S . As shown above, $-w$ is then an upper bound for S' , so $u = \sup S' \leq -w$, which gives $w \leq -u$.

Thus $-u$ is a lower bound for S that is at least as large as every other lower bound, so $-u = \inf S$. ■

Example 2.5.2

$$S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$$

For every $n \in \mathbb{N}$, the binomial theorem gives

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \sum_{k=2}^n \binom{n}{k} \frac{1}{n^k} > 1.$$

Indeed, for $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}.$$

For $k \geq 2$, we have $k! \geq 2^{k-1}$, because each factor from 2 through k is at least 2. Therefore

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

Thus $1 < (1 + 1/n)^n < 3$ for all $n \in \mathbb{N}$.

2.5.1 The Archimedean Property

We now prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. The proof combines two ingredients:

- Completeness of $\mathbb{R}$: this is what lets us form $u = \sup \mathbb{N}$ in the first place. Note carefully that u here is the supremum of $\mathbb{N}$ *regarded as a subset of* $\mathbb{R}$

- completeness says nothing about whether u itself is a natural number, and indeed the entire argument below is designed to show that assuming $\mathbb{N}$ is bounded above (so that such a u exists) leads to a contradiction.
- The inductive property of $\mathbb{N}$ (that $\mathbb{N}$ is closed under $+1$): this is the purely arithmetic fact that lets us manufacture a natural number strictly larger than u , which is what breaks the supremum.

Definition 2.5.3 (Archimedean Property)

If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.

In symbols

$$(\forall x \in \mathbb{R})(\exists n_x \in \mathbb{N}) [x \leq n_x]$$

Idea

If $\mathbb{N}$ were bounded above, let $u = \sup \mathbb{N}$. Then $u - 1$ is not an upper bound (the ϵ -characterization with $\epsilon = 1$), so some $m \in \mathbb{N}$ has $m > u - 1$ — and then $m + 1 \in \mathbb{N}$ lies beyond u .

Proof

Suppose, for contradiction, that $(\exists x \in \mathbb{R})(\forall n \in \mathbb{N}) [x > n]$.

Then x is an upper bound for $\mathbb{N}$. Since $1 \in \mathbb{N}$, $\mathbb{N} \neq \emptyset$, so by the Completeness Axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$.

Say $u = \sup \mathbb{N}$. Since

$$u - 1 < u,$$

and u is the *least* upper bound of $\mathbb{N}$, no number strictly smaller than u can be an upper bound for $\mathbb{N}$ (an upper bound smaller than the least upper bound would contradict its minimality). Applying this general principle to $u - 1$, we conclude that $u - 1$ is not an upper bound for $\mathbb{N}$. Hence

$$(\exists m \in \mathbb{N}) [u - 1 < m],$$

which gives

$$u < m + 1.$$

But $m + 1 \in \mathbb{N}$ (since $\mathbb{N}$ is closed under $+1$), and u is an upper bound for $\mathbb{N}$, so $m + 1 \leq u$. This contradicts $u < m + 1$: we have simultaneously $u < m + 1$ and $m + 1 \leq u$.

Therefore no such x exists; that is, for every $x \in \mathbb{R}$ there exists $n_x \in \mathbb{N}$ with $x \leq n_x$. ■

Completeness + closure of $\mathbb{N}$ under $+ 1 \implies$ Archimedean Property

Corollary 2.5.4

The following are consequences of the Archimedean Property:

(a) If

$$S = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\},$$

then

$$\inf S = 0.$$

 (b) If $t > 0$, there exists $n_t \in \mathbb{N}$ such that

$$0 < \frac{1}{n_t} < t.$$

 (c) If $y > 0$, there exists $n_y \in \mathbb{N}$ such that

$$n_y - 1 \leq y < n_y.$$

In symbols

$$(a) \quad \inf \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} = 0$$

$$(b) \quad (\forall t > 0)(\exists n_t \in \mathbb{N}) \left[0 < \frac{1}{n_t} < t \right]$$

$$(c) \quad (\forall y > 0)(\exists n_y \in \mathbb{N}) [n_y - 1 \leq y < n_y]$$

Idea

Each part is one move.

(a) 0 is a lower bound, and (b) says nothing larger is.

 (b) Archimedes applied to $1/t$.

 (c) Take the *least* natural number to the right of y : $n_y = \min \{n \in \mathbb{N} : y < n\}$ (well-ordering, not a supremum).

Proof

 (a) Clearly 0 is a lower bound for S . Let $b > 0$. By part (b), there exists $n \in \mathbb{N}$ such that $1/n < b$. Since $1/n \in S$, this means b is not a lower bound for S . As $b > 0$ was arbitrary, no number greater than 0 is a lower bound for S . Combined with 0 being a lower bound, $\inf S = 0$.

 (b) By the Archimedean Property, choose $m \in \mathbb{N}$ with $1/t \leq m$. Then $n_t = m + 1 \in \mathbb{N}$ satisfies $n_t > m \geq 1/t$, so $n_t > 1/t$, and dividing by

$n_t > 0$ gives $0 < 1/n_t < t$.

(c) Let

$$A = \{n \in \mathbb{N} : y < n\}.$$

By the Archimedean Property, A is nonempty (some natural number exceeds y). Since $A \subseteq \mathbb{N}$ is nonempty, the Well-Ordering Principle guarantees that A has a least element; let $n_y = \min A$. Then $n_y \in A$, so $y < n_y$.

It remains to show $n_y - 1 \leq y$. Suppose instead that $n_y - 1 > y$. Since $y > 0$ and $n_y \geq 1$ (as $n_y \in \mathbb{N}$), if $n_y = 1$ then $n_y - 1 = 0 > y$ is impossible because $y > 0$; so $n_y \geq 2$ and $n_y - 1 \in \mathbb{N}$. Then $n_y - 1 \in \mathbb{N}$ together with $n_y - 1 > y$ shows $n_y - 1 \in A$. But $n_y - 1 < n_y$, contradicting that n_y is the *least* element of A . Hence $n_y - 1 \leq y$, and combined with $y < n_y$ we obtain

$$n_y - 1 \leq y < n_y.$$

■

Remark

Part (c) is a *minimum* argument, not a supremum argument, and this distinction matters. The set $A = \{n \in \mathbb{N} : y < n\}$ is a subset of $\mathbb{N}$, and in general A is *unbounded above* in $\mathbb{R}$: if y is not itself a boundary case, every sufficiently large natural number also lies in A . So $\sup A$ would not even be a finite real number, and asking for $\sup A$ would be the wrong tool here. What we actually need is the *least* element of A , and the Well-Ordering Principle supplies this regardless of whether A is bounded above — it only requires A to be a nonempty subset of $\mathbb{N}$. This is in direct contrast to the supremum arguments used elsewhere in this section (for $\sup \mathbb{N}$ in the Archimedean Property, and for $\sup S$ in the existence of $\sqrt{2}$ below), where the relevant sets are subsets of $\mathbb{R}$ and boundedness above is essential.

2.5.2 The Existence of $\sqrt{2}$

We know $\sqrt{2} \notin \mathbb{Q}$. Does $\mathbb{R}$ contain a square root of 2? Completeness should produce one. Take all the candidates from below,

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

Its supremum $x = \sup S$ ought to sit exactly on the boundary between squares below 2 and squares above 2. There are only two ways x could miss:

- $x^2 < 2$: then x has not reached the boundary, so it can be nudged slightly *right* and still lie in S — then x is not an upper bound;
- $x^2 > 2$: then x has overshoot, so it can be nudged slightly *left* and still be an upper bound — then x is not the *least* one.

Unlike part (c) above, this is a genuine completeness argument: S is a bounded set of reals and we need its least upper bound, not a least element.

Proposition 2.5.5

There exists a real number x such that $x^2 = 2$.

In symbols

$$(\exists x \in \mathbb{R}) [x^2 = 2]$$

Scratch work

How small must the nudge $1/n$ be? Work backwards from what we want.

- *Right nudge.* Want $\left(x + \frac{1}{n}\right)^2 < 2$. Expanding gives $x^2 + \frac{2x}{n} + \frac{1}{n^2}$; the $\frac{1}{n^2}$ is awkward, but $\frac{1}{n^2} \leq \frac{1}{n}$, so it is enough that

$$x^2 + \frac{2x+1}{n} < 2 \iff n > \frac{2x+1}{2-x^2}.$$

- *Left nudge.* Want $\left(x - \frac{1}{n}\right)^2 > 2$. Dropping the positive $\frac{1}{n^2}$, it is enough that

$$x^2 - \frac{2x}{n} > 2 \iff n > \frac{2x}{x^2 - 2}.$$

Both thresholds are fixed positive numbers, so the Archimedean Property supplies such an n .

Proof

Define

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

We have $1 \in S$, so $S \neq \emptyset$, and 2 is an upper bound for S : if $t \in S$ and $t > 2$, then $t^2 > 4 > 2$, a contradiction. By the Completeness Axiom, S has a supremum in $\mathbb{R}$.

Let $x = \sup S$. Note $x \geq 1 > 0$. We show $x^2 = 2$ by ruling out $x^2 < 2$ and $x^2 > 2$.

Case 1: $x^2 < 2$. We find $n \in \mathbb{N}$ large enough that $x + \frac{1}{n} \in S$, contradicting that x is an upper bound for S .

Step 1 (bound the perturbed square). Since $x \geq 1$ and $n \geq 1$,

$$\begin{aligned}\left(x + \frac{1}{n}\right)^2 &= x^2 + \frac{2x}{n} + \frac{1}{n^2} \\ &\leq x^2 + \frac{2x}{n} + \frac{1}{n} \\ &= x^2 + \frac{2x+1}{n}.\end{aligned}$$

Step 2 (choose n via the Archimedean Property). Since $x^2 < 2$, the quantity $2 - x^2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x+1}{n} < 2 - x^2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x + \frac{1}{n}\right)^2 < x^2 + (2 - x^2) = 2,$$

so $x + \frac{1}{n} \in S$. But $x + \frac{1}{n} > x = \sup S$, contradicting that x is an upper bound for S . Hence $x^2 \geq 2$.

$$x^2 < 2 \implies x \text{ can be moved slightly to the right and remain in } S \implies x \neq \sup S.$$

Case 2: $x^2 > 2$. We find $n \in \mathbb{N}$ large enough that $x - \frac{1}{n}$ is an upper bound for S smaller than x , contradicting that $x = \sup S$.

Step 1 (bound the perturbed square). Since $1/n^2 > 0$,

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n}.$$

Step 2 (choose n via the Archimedean Property). Since $x^2 > 2$, the quantity $x^2 - 2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x}{n} < x^2 - 2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x - \frac{1}{n}\right)^2 > x^2 - (x^2 - 2) = 2.$$

So for any $t \in S$, we have $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$. Also $x - 1/n \geq 0$ because $x \geq 1$ and $n \geq 1$. Since $u \mapsto u^2$ is strictly increasing on $[0, \infty)$, this implies $t < x - \frac{1}{n}$. Thus $x - \frac{1}{n}$ is an upper bound for S smaller than $x = \sup S$, contradicting that x is the *least* upper bound. Hence $x^2 \leq 2$.

$x^2 > 2 \implies x$ can be moved slightly to the left and remain an upper bound for $S \implies x \neq \sup S$.

Combining both cases, $x^2 = 2$. ■

2.5.3 Density of $\mathbb{Q}$ and of the Irrationals

Completeness is exactly what $\mathbb{Q}$ is missing. Take

$$T = \{r \in \mathbb{Q} : r^2 < 2\},$$

the rational version of the set S above. It is nonempty ($0 \in T$) and bounded above in $\mathbb{Q}$ (every $r \in T$ has $r < 2$). Viewed inside $\mathbb{R}$, it has a supremum $y = \sup_{\mathbb{R}} T$, and the same two-case argument as above gives $y^2 = 2$, so $y = \sqrt{2}$. But $\sqrt{2} \notin \mathbb{Q}$, and y was the only possible least upper bound, so T has no supremum in $\mathbb{Q}$. Thus $\mathbb{Q}$ is *not* complete: $\mathbb{R}$ needs irrationals like $\sqrt{2}$ to serve as suprema of sets like T .

Still, $\mathbb{Q}$ is spread densely through $\mathbb{R}$. The picture: lay down the multiples $\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots$ on the number line. If the step $\frac{1}{n}$ is shorter than the interval (x, y) , the multiples cannot jump over it, so one of them lands inside.

Theorem 2.5.6

If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

In symbols

$$(\forall x, y \in \mathbb{R}) [x < y \implies (\exists r \in \mathbb{Q}) [x < r < y]]$$

Scratch work

Take $0 < x < y$ first. We want m, n with

$$x < \frac{m}{n} < y \iff xn < m < yn.$$

- Choose m to be the *first* natural number past xn (part (c)): then $xn < m \leq xn + 1$.
- This m stays below yn provided $xn + 1 < yn$, i.e. $\frac{1}{n} < y - x$ — which Archimedes provides.

For $x \leq 0$, shift both endpoints right by an integer $k > -x$, apply the positive case, and shift back.

Proof

First suppose that $0 < x < y$. Then $y - x > 0$. By the Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < y - x.$$

Multiplying $1/n < y - x$ by $n > 0$ shows that

$$xn + 1 < yn.$$

We now find a natural number between xn and $xn + 1$. Since $xn > 0$, part (c) of the preceding corollary (with y there replaced by xn) gives some $m \in \mathbb{N}$ with

$$m - 1 \leq xn < m.$$

Equivalently,

$$xn < m \leq xn + 1.$$

Combining this with $xn + 1 < yn$ obtained above, we have

$$xn < m < yn,$$

where the left inequality is immediate from $m > xn$, and the right inequality follows since $m \leq xn + 1 < yn$. Dividing throughout by $n > 0$, we get

$$x < \frac{m}{n} < y.$$

Since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, $m/n \in \mathbb{Q}$. This proves the theorem when $0 < x < y$.

Now suppose that $x \leq 0$. Choose $k \in \mathbb{N}$ such that $-x < k$ (possible by the Archimedean Property). Then adding k to $x < y$ and to $-x < k$ gives

$$0 < x + k < y + k.$$

By the positive case just proved, there exists $q \in \mathbb{Q}$ such that $x + k < q < y + k$. Let $r = q - k$. Since $q \in \mathbb{Q}$ and $k \in \mathbb{N} \subseteq \mathbb{Q}$, we have $r \in \mathbb{Q}$, and subtracting k from each part of $x + k < q < y + k$ gives

$$x < r < y.$$

■

Corollary 2.5.7

If x and y are any real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.

In symbols

$$(\forall x, y \in \mathbb{R}) [x < y \Rightarrow (\exists z \in \mathbb{R} \setminus \mathbb{Q}) [x < z < y]]$$

Idea

Scale by $\sqrt{2}$: a nonzero rational r between $x/\sqrt{2}$ and $y/\sqrt{2}$ gives $\sqrt{2}r$ between x and y , and $\sqrt{2}r$ is irrational.

Proof

Set $a = x/\sqrt{2}$ and $b = y/\sqrt{2}$. Since $\sqrt{2} > 0$, we have $a < b$. Choose a nonzero rational number r with $a < r < b$: if $b \leq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; if $a \geq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; and if $a < 0 < b$, choose $r \in \mathbb{Q}$ with $0 < r < b$. In each case, the density theorem provides such an r , and $r \neq 0$.

Let $z = \sqrt{2}r$. Multiplying $a < r < b$ by $\sqrt{2}$ gives $x < z < y$. If z were rational, then $\sqrt{2} = z/r$ would be rational, a contradiction. Thus z is irrational. ■

Corollary 2.5.8

Let $x, y \in \mathbb{R}$, with $x < y$. Then there exists infinitely many rationals between x and y .

In symbols

$$\begin{aligned} &(\forall x, y \in \mathbb{R}) [x < y \\ &\Rightarrow (\forall n \in \mathbb{N})(\exists r_1, \dots, r_n \in \mathbb{Q}) [x < r_1 < r_2 < \dots < r_n < y]] \end{aligned}$$

Proof

Choose $r_1 \in \mathbb{Q}$ with $x < r_1 < y$. Having chosen $r_k \in \mathbb{Q}$ with $x < r_k < y$, apply the density theorem to $r_k < y$ to choose $r_{k+1} \in \mathbb{Q}$ such that

$$r_k < r_{k+1} < y.$$

This produces infinitely many distinct rational numbers $r_1, r_2, \dots$ in (x, y) . ■

2.5.4 Countable and Uncountable Sets

Are there more rational or irrational numbers? To answer this question, we need to know in what sense.

Definition 2.5.9 (Cardinality)

A function $f : A \rightarrow B$ is a *bijection* if it is both injective (one-to-one) and surjective (onto). We say that A and B have the same *cardinality*, written

$$|A| = |B|,$$

if there exists a bijection $f : A \rightarrow B$. Thus, cardinality records the size of a set up to a bijection.

Definition 2.5.10

A set S is *finite* if $S = \emptyset$ or if there is a bijection

$$\{1, 2, \dots, n\} \rightarrow S$$

for some $n \in \mathbb{N}$. A set S is *countably infinite* if there is a bijection $f : \mathbb{N} \rightarrow S$.

Example 2.5.11 (a) The function $f : \mathbb{N} \rightarrow \{2, 4, 6, \dots\}$ defined by $f(n) = 2n$ is a bijection. Thus the set of even natural numbers has the same cardinality as $\mathbb{N}$.

(b) The function $g : \mathbb{N} \rightarrow \mathbb{Z}$ defined by

$$g(1) = 0, \quad g(2k) = k, \quad g(2k + 1) = -k \quad (k \in \mathbb{N})$$

is a bijection. Thus $\mathbb{Z}$ has the same cardinality as $\mathbb{N}$.

Definition 2.5.12

A set S is *countable* if it is finite or countably infinite. A set is *uncountable* if it is not countable.

3 Sequences and Limits

3.1 Convergent Sequences

3.1.1 The Limit of a Sequence

A sequence of real numbers is a function $X : \mathbb{N} \rightarrow \mathbb{R}$. We write $x_n = X(n)$ for its n th term and denote the sequence by $\{x_n\}_{n=1}^{\infty}$.

Definition 3.1.1

A sequence $\{x_n\}$ in $\mathbb{R}$ converges to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there is an index $n_\epsilon \in \mathbb{N}$ such that

$$n \geq n_\epsilon \implies |x_n - a| < \epsilon.$$

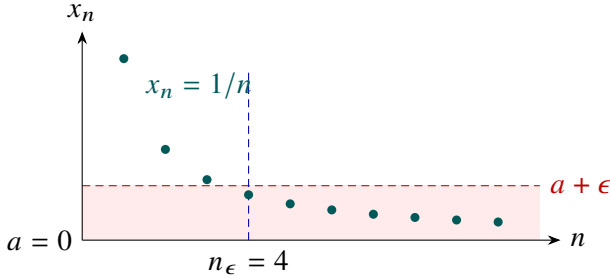
We then write $\lim_{n \rightarrow \infty} x_n = a$.

In symbols,

$$x_n \rightarrow a \iff (\forall \epsilon > 0)(\exists n_\epsilon \in \mathbb{N})(\forall n \geq n_\epsilon) [|x_n - a| < \epsilon].$$

In words: for every tolerance $\epsilon > 0$, all sufficiently late terms lie in $(a - \epsilon, a + \epsilon)$. The index n_ϵ may depend on ϵ , and once $n \geq n_\epsilon$ the inequality must hold for every later term, even if earlier terms lie outside the interval.

The points below show $x_n = 1/n$. For the illustrated tolerance $\epsilon = 0.3$, every term from $n_\epsilon = 4$ onward lies in the shaded band around $a = 0$.



For $a \in \mathbb{R}$, the ϵ -neighbourhood of a is

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = (a - \epsilon, a + \epsilon).$$

Thus $x_n \rightarrow a$ precisely when, for every $\epsilon > 0$, there is $n_\epsilon \in \mathbb{N}$ such that $x_n \in V_\epsilon(a)$ for all $n \geq n_\epsilon$.

Example 3.1.2

Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Proof

Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Hence $1/n \rightarrow 0$. ■

Example 3.1.3

Prove that $\lim_{n \rightarrow \infty} \frac{4n+3}{n+1} = 4$.

Scratch work

Simplify the distance first, then see how large n must be:

$$\left| \frac{4n+3}{n+1} - 4 \right| = \left| \frac{-1}{n+1} \right| = \frac{1}{n+1} < \frac{1}{n}.$$

So it is enough that $\frac{1}{n} < \epsilon$. By Archimedes pick n_ϵ with $\frac{1}{n_\epsilon} < \epsilon$; every $n \geq n_\epsilon$ then works.

Proof

Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Therefore the sequence converges to 4. ■

3.1.2 Uniqueness of the Limit

Proposition 3.1.4

If $x, y \in \mathbb{R}$ and $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

In symbols

$$(\forall x, y \in \mathbb{R}) \left[(\forall \epsilon > 0) [|x - y| < \epsilon] \Rightarrow x = y \right]$$

Proof

Suppose $x \neq y$. Then $\epsilon = |x - y|/2 > 0$, but $|x - y| > \epsilon$, contradicting the hypothesis. ■

Theorem 3.1.5

A sequence in $\mathbb{R}$ can have at most one limit.

In symbols

$$(\forall a, b \in \mathbb{R}) [(x_n \rightarrow a \wedge x_n \rightarrow b) \Rightarrow a = b]$$

Scratch work

By the preceding proposition, it suffices to show $|a - b| < \epsilon$ for every $\epsilon > 0$. The only link between a and b is the sequence, so pass through a term x_n :

$$|a - b| \leq |a - x_n| + |x_n - b|.$$

To make the sum $< \epsilon$, make each piece $< \epsilon/2$: that needs $n \geq n_1$ for the first and $n \geq n_2$ for the second, so take $n \geq \max \{n_1, n_2\}$.

Proof

Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Given $\epsilon > 0$, choose $n_1, n_2 \in \mathbb{N}$ so that $|x_n - a| < \epsilon/2$ whenever $n \geq n_1$ and $|x_n - b| < \epsilon/2$ whenever $n \geq n_2$. For $n \geq \max \{n_1, n_2\}$, the triangle inequality gives

$$|a - b| \leq |a - x_n| + |x_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This holds for every $\epsilon > 0$, so the preceding proposition gives $a = b$. ■

3.1.3 Divergence and Boundedness

A sequence diverges if it does not converge to any real number. For example, consider $\{x_n\} = \{0, 1, 0, 1, \dots\}$, with $x_n = 0$ for odd n and $x_n = 1$ for even n .

Both 0 and 1 keep appearing arbitrarily late, so the terms cannot all remain close to one proposed limit.

Suppose $x_n \rightarrow a$ and take $\epsilon = 1/2$. There is an n_ϵ such that $|x_n - a| < 1/2$ for every $n \geq n_\epsilon$. Choose an odd and an even index beyond n_ϵ . The odd term gives $|a| < 1/2$, hence $-1/2 < a < 1/2$. The even term gives $|1 - a| < 1/2$, hence $1/2 < a < 3/2$. These intervals do not overlap, a contradiction. Thus this sequence diverges.

Definition 3.1.6

A sequence $\{x_n\}$ is bounded if there is an $M \in \mathbb{R}$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 3.1.7

Every convergent sequence of real numbers is bounded.

In symbols

$$(\exists a \in \mathbb{R}) [x_n \rightarrow a] \Rightarrow (\exists M \in \mathbb{R})(\forall n \in \mathbb{N}) [|x_n| \leq M]$$

Idea

The tail stays within 1 of the limit; only finitely many terms come before it, and a finite set has a maximum.

Proof

Suppose $x_n \rightarrow a$. Taking $\epsilon = 1$, choose $n_1 \in \mathbb{N}$ so that $|x_n - a| < 1$ for all $n \geq n_1$. For these terms, the triangle inequality gives

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

The remaining terms form a finite set. Thus

$$M = \max \{|x_1|, \dots, |x_{n_1-1}|, 1 + |a|\}$$

bounds every term of the sequence (when $n_1 = 1$, take $M = 1 + |a|$). ■

The converse fails: $\{0, 1, 0, 1, \dots\}$ is bounded but, as shown above, diverges.

Example 3.1.8

Consider the following sequences.

$$- \{a_n\} = \left\{ \frac{n^2+1}{n} \right\}$$

Take a look at this sequence:

$$\left| \frac{n^2+1}{n} \right| = \left| n + \frac{1}{n} \right| = n + \frac{1}{n}$$

For any $M > 0$, there exist $n_m \in \mathbb{N}$ such that $n_m > M$. Thus,

$$|a_{n_m}| > M.$$

$$- \{a_n\} = \{(-1)^n n\}$$

Take a look at this sequence:

$$|a_n| = |(-1)^n n| = |(-1)^n| |n| = |n| > M$$

3.1.4 Divergence to $\pm\infty$

Definition 3.1.9

A sequence $\{x_n\}$ diverges to $+\infty$, that is, $\lim_{n \rightarrow \infty} x_n = \infty$, if for all real number $M > 0$, there exists $n_M \in \mathbb{N}$ such that $a_n > M$ for all $n \geq n_M$.

Definition 3.1.10

A sequence $\{x_n\}$ diverges to $-\infty$, that is, $\lim_{n \rightarrow \infty} x_n = -\infty$, if for all real number $M > 0$, there exists $n_M \in \mathbb{N}$ such that $a_n < -M$ for all $n \geq n_M$.

We write $\lim_{n \rightarrow \infty} x_n = \pm\infty$, but such a sequence is still divergent: it does not converge to any real number.

Example 3.1.11

Prove that $\lim_{n \rightarrow \infty} \frac{n^2}{1-n} = -\infty$.

Proof

Let $M > 0$.

■

Example 3.1.12

Prove that $\lim_{n \rightarrow \infty} \sqrt{n} + 1 = +\infty$.

Proof

The goal is to let $\sqrt{n} > m - 1$

■

Theorem 3.1.13

If $\lim_{n \rightarrow \infty} x_n = a$ and $\lim_{n \rightarrow \infty} y_n = b$, then

- (i) $(\forall c \in \mathbb{R}) [\lim_{n \rightarrow \infty} c = c]$
- (ii) $(\forall c \in \mathbb{R}) [\lim_{n \rightarrow \infty} cx_n = ca]$

In symbols

$$(x_n \rightarrow a \wedge y_n \rightarrow b) \Rightarrow (\forall c \in \mathbb{R}) \left[\lim_{n \rightarrow \infty} c = c \wedge \lim_{n \rightarrow \infty} cx_n = ca \right]$$